The Role of Mutuality and Coping in a Nurse-Led Cognitive Behavioral Intervention on Depressive Symptoms Among Dementia Caregivers

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ABSTRACT

The current study explored whether mutuality and coping predicted and/or mediated the effect of a nurse-led cognitive behavioral intervention (NLCBI) on depressive symptoms of caregivers of persons with dementia. The intervention group (n = 56) received five monthly in-home nurse-led cognitive behavioral sessions and consultation calls after each session. The control group (n = 56) received five monthly short general conversations with the nurse interventionist. Questionnaires on study variables and demographics were collected at baseline, end of intervention, and 2-month follow up. Improved mutuality (β = –0.75, p = 0.049) and active coping (β = –2.06, p = 0.0001) and decreased passive coping (β = 1.43, p = 0.001) were found to predict the reduction of depressive symptoms among caregivers in the NLCBI. However, none of these variables mediated the interventional effect. Regular mental health nursing interventions are suggested to focus on enhancing mutuality and active coping and decreasing passive coping to maintain caregivers' mental health.

Targets: Caregivers of persons with dementia.

Intervention Description: Nurse-led cognitive behavioral sessions and subsequent consultation calls.

Mechanisms of Action: Impacted caregivers' reappraisals, thus improving their active coping skills and mutuality and decreasing their passive coping, which directly reduced their depressive symptoms.

Outcomes: Mutuality, active coping, and passive coping played a predicting, but not mediating, role in the effect of the NLCBI.

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Family caregivers of persons with dementia (PWD) often experience depressive symptoms, as dementia care is burdensome and of long duration. Caregiving situations are relatively confined as caregivers need to closely attend their PWD and may become homebound, so dyadic mutuality, a positive quality of the caregiver–care recipient relationship, is important to the mental health of family caregivers (Park & Schumacher, 2014). However, inter- and/or intrapersonal conflicts may arise amidst the caregiving process, damaging mutuality. In China, although caregiving for PWD is

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primarily a family responsibility according to cultural tradition and legislation, the familial network has diminished with urbanization and the previous family planning policy, and caregiver support systems are still underdeveloped. Thus, family caregivers need effective coping strategies to maintain their mental health.

THEORETICAL FRAMEWORK

Lazarus and Folkman's (1984) Stress and Coping Theory was used to guide the design of the current study. Stress and Coping Theory emphasizes the reciprocal relationship between an individual and his/her surroundings. Stressors are seen as deriving from an imbalance between perceived demands and resources, and *coping* refers to an individual's cognitive and behavioral efforts in managing demands that are appraised as taxing or exceeding an individual's resources (Lazarus & Folkman, 1984). In the process of coping, primary and secondary appraisals converge to determine whether the person–environment transaction is significant for well-being, whereas reappraisals occur based on new information. Lazarus and Folkman (1984) posit that cognitive appraisals in person–environment transactions impact an individual's coping behaviors and health outcomes.

Psychosocial programs such as cognitive behavioral interventions (CBIs) may help adjust caregivers' reappraisals on dementia, role demands, and dyadic relationships within specific social contexts; caregivers' interactions with these situations could be adapted using more positive emotions and active coping skills. Therefore, positive reappraisals within transactions may improve caregivers' coping strategies and dyadic relationships, and consequently promote caregivers' mental health (Zauszniewski, Lekhak, & Musil, 2018).

Mutuality and Coping in the Interventional Effect on Caregiver Depression

The Role of Mutuality. Mutuality, a positive quality of dyadic relationships, was viewed as a support resource for caregivers' mental health (Pan, Jones, & Winslow, 2016). Mutuality has four dimensions: affection, reciprocity, sharing of pleasant activities, and sharing of values (Archbold, Stewart, Greenlick, & Harvath, 1990). Extensive literature indicates that higher mutuality is associated with less burden or role strain, fewer depressive symptoms, and moderation or mediation between stress/caregiving demands and mental health among family caregivers (Crist, Pasvogel, Szalacha, & Finley, 2017; Park & Schumacher, 2014; Yang, Liu, & Shyu, 2014). However, few studies have been conducted using supportive strategies such as CBIs to facilitate caregivers' mutuality, and some studies with relevant interventions revealed inconsistent effects on mutuality or depression.

Kunik et al. (2017) reported that mutuality, but not depression, improved among dementia caregivers with strategies of enhancing communication and making daily activities pleasant. However, Li, Powers, et al. (2012) did not find significant group differences in either mutuality or depression, although caregivers with an empowerment educational program had less role strain during follow up. Gonzalez, Polansky, Lippa, Gitlin, and Zauszniewski (2014) further reported little effect on these variables after a resourcefulness training program among caregivers of PWD. These studies included both mutuality and depression as dependent variables; thus, the role of mutuality in interventions on caregiver depression is unknown (Rausch, Caljouw, & Es, 2017).

The Role of Coping. Coping is a process in which an individual mobilizes his/her resources in response to demands. Coping strategies vary among individuals and situations but are often categorized as active and passive coping, or problem-focused and emotion-focused coping (Lazarus & Folkman, 1984; Xie, 1998). *Active coping* includes positive emotions and problem-solving strategies, such as finding solutions and seeking help, whereas *passive coping*, similar to dysfunctional coping, includes negative emotions such as avoidance and denial, and behaviors such as substance use and daydreaming (Li, Cooper, Bradley, Shulman, & Livingston, 2012; Roche, MacCann, & Croot, 2016). Evidence showed that more active coping reduced depressive symptoms among caregivers, whereas more passive/dysfunctional coping increased depressive symptoms (Chan & Chui, 2011; Lu, Liu, Wang, & Lou, 2017; Roche, Croot, MacCann, Cramer, & Diehl-Schmid, 2015).

Several studies have explored the effect of cognitive-behavioral therapy (CBT) on caregivers' coping strategies, but results varied. Studies concluded three different results: (a) active coping strategies improved (Fialho, Koenig, dos Santos, Barbosa, & Caramelli, 2012; Mahmoodi, Khani, Banab, & Bagheri, 2016); (b) positive and dysfunctional coping strategies increased (Li, Cooper, Austin, & Livingston, 2013); and (c) active coping strategies increased, whereas passive coping strategies selectively decreased (Losada, Marquez-Gonzalez, & Romero-Moreno, 2011). These inconsistencies raised the question of whether coping strategies play any role in the effect of CBT on caregivers' depressive symptoms.

In addition, findings from several relevant studies were conflicted on how CBT relieved caregiver depression. Gallagher-Thompson, Gray, Dupart, Jimenez, and Thompson's (2008) report showed that CBT worked via skill utilization and helpfulness among non-Hispanic White and Hispanic-Latino dementia caregivers. However, Li et al. (2014) found that the effect of a coping skill intervention was mediated by increased emotion-focused coping among caregivers with higher baseline depression, with neither problem-focused nor dysfunctional coping playing a role. A similar study by Losada et al. (2011) indicated that this effect occurred by modifying caregivers' dysfunctional thoughts. Therefore, as caregivers may respond to interventions differently, more research needs to be conducted to elucidate whether coping is a mediator in successful CBT interventions.

PURPOSE

Little evidence exists regarding the role of mutuality and coping in the effect of nurse-led psychosocial programs on caregivers' mental health. The purpose of the current study was to explore whether mutuality and coping either predict or mediate the effects of a nurse-led CBI (NLCBI) on depressive symptoms of family caregivers of PWD. Nursing strategies and social policies can be developed to maintain caregivers' mental health in countries with similar care burdens and socioeconomic contexts such as those in China.

METHOD

Study Design and Sample

This study was a randomized controlled trial approved by Jin Hua Science and Technology Bureau, including review on human subjects. Family caregivers were recruited via resident and staff referrals from community health centers and neurological units of a tertiary hospital of a city in southeast China. Caregivers were recruited if they: (a) were a primary caregiver; (b) were age 20 or older; (c) were providing care to a family member diagnosed with Alzheimer's disease or dementia (including vascular dementia), or with a Mini-Mental State Examination (MMSE) score ≤17; (d) were living in communities of the city or surrounding areas; (e) had a depression score on the Center for Epidemiologic Studies Depression Scale (CES-D) of ≥10 and <20, for better response to the intervention and elimination of medical treatments or therapies; and (f) were able to speak or understand Chinese (Putonghua). Respondents were excluded, as interviewed and screened with medical history and records by the researcher, if they had suicidal tendencies, a history of psychotic disorders, signs of intellectual deficits, and/or were undergoing psychosocial therapies and medication treatments.

Sample Size

G*power achieved a sample size of 98 with an *F* test, three repeated measurements for two independent groups, effect size = 0.25, power = 0.85, and expected r = 0.5 between measurements. Effect size was decided based on a systematic review and meta-analysis that showed a small-to-medium effect of CBT on caregiver depression (Liu, 2017). Whereas most psychosocial studies had an attrition rate of 5% to 30% (Cheng et al., 2017; Losada et al., 2015), the current study used a 15% attrition rate, according to Waldorff et al.'s (2012) 11% to 20% attrition rate at 6- to 12-month follow up. Thus, the sample size was finalized to 112 individuals.

Procedure

Entry permissions of the project to research sites were obtained from authorities of the community health centers and hospital. Flyers and word of mouth were used to explain the project and its recruitment criteria. The researcher (Y.P.) contacted potential respondents after referrals to confirm eligibility, obtain verbal consent, and help arrange an appointment for the baseline survey. Five trained research assistants conducted face-to-face surveys anonymously at Time 1 (*baseline*), Time 2 (*end of the 5-month intervention*), and Time 3 (*2-month follow up*) at caregivers' homes or in a hospital unit, as convenient. Research assistants were trained with survey procedures and interview skills, and were provided answers for possible questions. They were qualified to conduct the survey after successfully demonstrating the procedures as supervised by the researcher. Caregivers were asked to fill out questionnaires, and research assistants helped caregivers read the questionnaire if needed. Patients' MMSE was assessed by research assistants.

Questionnaires were collected upon completion. After the initial assessment, caregivers were randomly allocated by a data processor into either the intervention or control group via a computerized random number, with 56 caregivers in each group. The researcher was initially blinded to the study allocation but supervised the intervention; data collectors were blinded to the treatment assignments; and caregivers received a sealed letter with an allocated number and were informed of their treatment when the intervention began. Caregivers were assigned to one of the five trained nurse interventionists by geographic location. Masking was not possible for the nurse interventionists. Collected data were locked in the researcher's office drawer, and the data file was encrypted and stored on the researcher's personal computer.

Cognitive Behavioral Intervention Group and Control Group

Intervention Group. This NLCBI comprised five monthly 60-minute individual sessions conducted at caregivers' homes at the beginning of each month by appointment, with a 20- to 30-minute consultation call in the middle of the month. Duration of the intervention and follow up was determined based on the time of an interventional effect and retention of respondents. Five RNs were trained on the intervention protocol and psychosocial communication techniques. Although nurses were instructed to follow the protocol, minimal flexibility was allowed to adapt to individual caregivers. A discussion group in an E-chat room using WeChat software was set up for communication among nurse interventionists to ensure consistency in the intervention. At the initial session, caregivers agreed to make up a session within a few days after a missed appointment.

Content of the intervention sessions was constructed and tailored to Chinese caregivers based on the cognitive behavioral strategies introduced by Aboulafia-Brakha, Suchecki, Gouveia-Paulino, Nitrini, and Ptak (2014); Cheng et al. (2017); and Schinköthe and Wilz (2014). Session 1 assessed caregivers' situations and perceptions on caregiving-related issues. Session 2 focused on clarifying caregivers' stressors, appraisals, and coping strategies. Session 3 primarily concentrated on cognitive training by teaching caregivers positive reappraisal strategies, particularly benefit-finding (i.e., viewing challenging situations in positive ways). In Session 4, caregivers were trained in active coping behaviors such as seeking help and nurturing healthy relationships, as well as in avoiding passive coping behaviors. Session 5 provided caregivers with relaxation and self-maintenance skills. Sessions began by building rapport with the caregiver and ended by summarizing the session and assigning homework. Questions used in the face-to-face intervention sessions are listed in **Table 1**.

As mutually agreed, the 20- to 30-minute consultation call was arranged to assess caregivers' recent condition, ask for feedback, and reinforce strategies taught in the sessions. Interventionists were prepared to answer caregivers' questions. Although subsequent sessions still abided by the protocol, caregivers' feedback was sought to help further clarify details of the intervention. No retention incentives were used, as attrition could partly reflect the effectiveness of the intervention.

Control Group. Caregivers in the control group were provided with five monthly, short (5 to 10 minutes), general conversations with a nurse interventionist at the beginning of each month either at caregivers' homes, at the hospital, or via telephone as convenient. No in-depth instructions were given; conversations were casual chats on caregivers' daily lives and health.

Measures

Depression. The 10-item CES-D was used to assess frequency of depressive symptoms among caregivers. One example of an item was "I felt depressed." Items were rated on a 4-point response format from 0 (*rarely or none of the time*) to 3 (*most of the time or all of the time*). The scale score is the total of the item scores, with ≥10 indicating the presence of clinical depression. The Chinese version of the CES-D had a Cronbach's alpha of 0.79 among Chinese caregivers (Pan et al., 2016).

Mutuality. The 15-item Mutuality Scale was used to measure caregiver mutuality. One example of an item was "To what extent do you enjoy the time the two of you spend together?" Items were rated on a 5-point Likert scale ranging from 0 (*not at all*) to 4 (*a great deal*). The scale score is the mean of the item scores; higher scores indicate higher levels of mutuality. The Chinese version of the scale had a Cronbach's alpha of 0.91 among caregivers in China (Pan et al., 2016).

The Simplified Coping Scale. The 20-item Simplified Coping Scale (Xie, 1998) was developed to measure coping strategies of Chinese adults, with a Cronbach's alpha of 0.90 for the full scale, 0.89 for the 12-item subscale of active coping, and 0.78 for the eight-item subscale of passive coping. Examples of items included "Seeking suggestions from friends and relatives" and "Accepting the reality as there are no other choices." Items were rated on a 4-point scale ranging from 0 (*never used*) to 3 (*often used*). The subscale scores are the respective averages of their items. A higher score in active coping indicates better coping, whereas a higher score in passive coping indicates poorer coping.

Care Recipients' Activities of Daily Living. The 14-item Activities of Daily Living (ADL) scale measures the basic and more complicated activities of patients' self-maintenance. One example of an item was "Are you having difficulty eating?" Items were rated on a 4-point scale from 1 (*performed without difficulty*) to 4 (*unable to perform*).

TABLE 1

Key Questions Leading the Face-to-Face Sessions of the Cognitive Behavioral Intervention

Session Number and Objectives	Questions Asked
Session 1 Understand caregivers' situations and perceptions	1. Could you please introduce yourself and describe your caregiving background (socioeconomic status, caregiving history, family network, caregiving demand, and support)? 2. How do you perceive dementia, the person with dementia, your role as a dementia caregiver, your family or dyadic relationship, your available support, and your utilization of this support? 3. Have you felt or experienced the social stigma of dementia?
Session 2 Clarify caregivers' stressors, appraisals, and coping strategies	1. Could you please list a number of difficult caregiving scenarios, your previous management approaches, and consequences of these approaches? 2. Could you please reflect on these scenarios above and identify the stressors, your appraisals, and coping strategies? 3. Are you aware of the positive appraisals and active coping strategies that may have reduced your stress and promoted your problem-solving skills as well as negative appraisals and passive coping strategies that may have increased your stress?
Session 3 Elicit positive appraisals and avoid negative appraisals	1. Could you please recall a couple of the most pleasant or most unhappy caregiving scenarios, and discuss your positive or negative appraisals during the process? For example, do you usually view dementia care and the dyadic relationship as negative or positive? 2. Could you please analyze those pleasant caregiving scenarios and identify the use of the strategy of benefit-finding? For example, have you experienced benefits or gains during the process? 3. Could you please try to apply the strategy of benefit-finding into the unhappy caregiving scenarios, avoid negative emotions, and imagine if the consequences would have been better?
Session 4 Train caregivers to use active coping behaviors and avoid passive coping behaviors	1. Could you please explore the most pleasant or most unhappy caregiving scenarios further, and discuss your active or passive coping behaviors during the process? 2. Could you please try to apply active coping behaviors to the most difficult caregiving scenarios, such as building support, seeking help, and nurturing a good dyadic relationship or a friendly caregiving environment, and imagine if the outcomes would have been better? 3. Would you like to reinforce these active coping behaviors in other challenging caregiving scenarios, and also avoid the use of passive coping behaviors such as substance use and avoidance?
Session 5 Train caregivers on skills for relaxation and self-maintenance	1. What relaxation skills do you use? Please list relaxation skills that might be practical to you. 2. Could you please spend some time now to practice relaxation skills such as deep breathing, writing in a diary, listening to music, exercising, and throwing soft pillows? 3. Do you know how to manage your time, resume pleasant activities or hobbies, balance nutrients, and get better sleep? 4. What other personal skills could you develop that might be beneficial to your health?

The scale score was the sum of the item scores. The Chinese version of the 14-item ADL scale (ADL-C) was used among Chinese family caregivers, with a Cronbach's alpha of 0.97 (Pan et al., 2016).

Care Recipients' Mini Mental State Examination. The Chinese version of the MMSE was used to assess patients' cognitive function. This tool has been widely tested with satisfactory psychometrics and applied to older adults in China (Yu, Li, & Huang, 2012). The MMSE comprises 10 cognitive function domains, such as memory, orientation, and calculation, with a maximum of 30 points. Higher scores indicate better cognitive function. The cut-off for diagnosing cognitive impairment is 16 to 19 for individuals with primary education and below; thus, a 17-point cut-off was adopted in the current study (Yang et al., 2016).

Demographic and Caregiving Characteristics. A questionnaire designed by the researchers was used to collect demographic data of caregivers and care recipients.

Data Analysis

SPSS 25.0 was used for data analysis. The missForest method (Stekhoven & Bühlmann, 2012) was adopted to replace missing data, including some missing questions from individuals who completed the study and missing questionnaires from those who dropped out, so that the replaced data could best reflect the nature of the true data. Descriptive data were evaluated with frequencies or measures of central tendency. Baseline comparisons in demographics and studied variables between the intervention group and control group, and between individuals who completed the study and those who dropped out, were analyzed using chi-square test, independent sample *t* test, or Mann-Whitney U test, in accordance with the variable type and whether assumptions were met.

Generalized linear mixed model (GLMM) was used to analyze the effect of the NLCBI on caregivers' depressive symptoms, and the predicting role of mutuality, active coping, and passive coping in this interventional effect. This analysis allows both the fixed and random effects to be explored simultaneously. The repeated measures (Time 1, Time 2, and Time 3) of the studied variables were reconstructed. Group, time, the interaction of group and time, and the studied variables were input to the fixed effect model, considering the covariates of caregivers' gender and patients' ADL and MMSE scores based on their significance with baseline depression scores by a primary screen. The time variable was placed into the random model, and the reconstructed depression score was the dependent variable. β and p values were presented.

To explore the potential mediating effect of mutuality, active coping, and passive coping, Baron and Kenny's (1986) procedure was adopted as suggested by Murphy, Cooper, Hollon, and Fairbum (2009), and as applied in similar studies (Li, Cooper, et al., 2014; Losada et al., 2011). A mediator needs to meet the following conditions analyzed through respective regression analysis: (a) variations in the independent variable (intervention group) must account for variations in the mediator (mutuality, active coping, or passive coping); (b) variations in the independent variable must account for variations in the dependent variable (depressive symptoms); and (c) while regressing the dependent variable on both the independent variable and the mediator, the independent variable must reduce or even lose its effect on the dependent variable, while the mediator must sustain its significance in affecting the dependent variable. The changes of the studied variables were calculated respectively by subtracting Time 1 (T_1) scores from Time 2 (T_2) or Time 3 (T_3) scores. The same covariates as in the GLMM were controlled.

RESULTS

The current study was conducted from May 2016 to March 2017 in a city in southeast China. Family caregivers of PWD (N = 112; intervention group n = 56, control group n = 56) were recruited from the 276 caregivers referred and were evaluated at Time 1. At Time 2, 99 caregivers (intervention group n = 54, control group n = 45) completed the survey; and at Time 3, 82 caregivers (intervention group n = 47, control group n = 35) completed the full three evaluations. Among dropouts in the intervention group, two caregivers' PWD died, six caregivers discontinued caregiving, and one caregiver relocated; in the control group, seven caregivers' PWD died, three caregivers discontinued caregiving, eight caregivers relocated, and three caregivers refused the repeated surveys.

Demographics of family caregivers and PWD are listed in **Table 2**. No significant differences were found in baseline characteristics between the intervention group and control group. Means and standard deviations of the studied variables are listed in **Table 3**. Significant group differences were indicated by the independent sample *t* test in depressive symptoms and active coping at Time 2 and Time 3, respectively.

In **Table 4**, the fixed model of GLMM revealed that, after controlling for covariates, the NLCBI was effective in reducing caregivers' depressive symptoms over time (group and time interaction, β = 1.12, p = 0.0001). Group alone was almost significant (β = 1.41, p = 0.053) but time was highly significant with reducing depressive symptoms. Both active coping (β = –2.06, p = 0.0001) and passive coping (β = 1.43, p = 0.001) strongly predicted caregivers' depressive symptoms in this NLCBI, whereas mutuality was a weaker predictor (β = –0.75, p = 0.049).

As shown in **Table 5,** none of the T_2T_1 or T_3T_1 changes in active coping played a mediating role in the effect of the NLCBI on the corresponding changes of caregivers' depressive symptoms after covariates were adjusted. The significance of group (the independent variable) was not affected by adding the change of active coping (the potential mediator) to the models (active coping: T_2T_1 [p = 0.002], T_3T_1 [p = 0.028]; group: p < 0.0001 in both models). Mutuality and passive coping did not meet the first condition as a mediator. Group was not a significant variable in

TABLE 2

Participant Characteristics at Baseline

Variable	n (%)			Statistical Test (p Value)
	Total ($N = 112$)	Intervention Group ($n = 56$)	Control Group ($n = 56$)	
Caregivers				
Age (years) (mean [SD])	62.68 (10.86)	63.27 (11.22)	62.09 (10.56)	$t = 0.57$ (0.57)
Gender				$\chi^2 = 2.44$ (0.12)
Female	70 (62.50)	31 (55.36)	39 (69.64)	
Male	42 (37.50)	25 (44.64)	17 (30.36)	
Relationship to care recipient				$\chi^2 = 0.14$ (0.71)
Spouse	54 (48.21)	28 (50)	26 (46.43)	
Other family member	58 (51.79)	28 (50)	30 (53.57)	
Educational level				$\chi^2 = 0.001$ (0.99)
Middle school and below	100 (89.29)	50 (89.29)	50 (89.29)	
High school and above	12 (10.71)	6 (10.71)	6 (10.71)	
Number of chronic diseases (mean [SD])	1.08 (1.13)	1.31 (1.29)	0.86 (0.90)	$Z = -1.89$ (0.07)
Duration of care (months) (mean [SD])	67.25 (75.58)	71.2 (85.57)	63.27 (64.61)	$Z = -0.52$ (0.60)
Care recipients				
Age (years) (mean [SD])	79.48 (9.54)	79.00 (9.35)	79.96 (9.78)	$t = -0.53$ (0.60)
Gender				$\chi^2 = 3.12$ (0.08)
Female	71 (63.39)	40 (71.43)	31 (55.36)	
Male	41 (36.61)	16 (28.57)	25 (44.64)	
ADL score[a] (mean [SD])	43.91 (10.54)	42.93 (10.70)	44.89 (10.38)	$t = -0.98$ (0.33)
MMSE score[b] (mean [SD])	6.54 (5.87)	6.45 (5.87)	6.64 (5.91)	$t = -0.18$ (0.86)

Note. ADL = activities of daily living; MMSE = Mini Mental Status Examination. *Z* scores were calculated using a Mann-Whitney U test.
[a] Scores range from 14 to 56, with higher scores indicating lower ability to perform ADL.
[b] Scores range from 0 to 30, with higher scores indicating better cognitive function.

the respective model of T_2T_1 change in mutuality ($\beta = 0.13$, $p = 0.20$) and passive coping ($\beta = -0.04$, $p = 0.69$). The models of their T_3T_1 changes achieved similar results. Thus, further analyses were not necessary.

As the attrition rate was higher in the control group (37.5%) than in the intervention group (16%), comparison analyses between individuals who completed ($n = 82$) and dropped out of ($n = 30$) the intervention were performed in the key baseline characteristics of the dyads. No significant group differences were found in caregiver genders ($\chi^2 = 0.58$, $p = 0.49$), relationship to care recipient ($\chi^2 = 2.19$, $p = 0.20$), or depressive symptoms ($t = 0.57$, $p = 0.57$); or patients' age ($t = -0.23$, $p = 0.82$), gender ($\chi^2 = 0.01$, $p = 0.99$), ADL score (Mann-Whitney U, $p = 0.051$), and MMSE scores (Mann-Whitney U, $p = 0.70$). However, younger caregivers (mean age = 58.93 years, $SD = 9.08$ years) were more likely to drop out than older caregivers (mean age = 64.05 years, $SD = 11.18$ years; $t = -2.25$, $p = 0.037$).

DISCUSSION

Consistent with findings from the literature (Hopkinson, Reavell, Lane, & Mallikarjun, 2018; Liu, 2017), this NLCBI was found to be therapeutic for caregiver depression, as the favored changes in caregivers' mutuality and coping played a key role. Because few similar studies have been conducted by nurses in the target population, this finding enriches the field of research and informs authorities that CBIs by community

TABLE 3

Means, Standard Deviations, and *t* Scores of Studied Variables Over Time

Variable	Mean (*SD*)		
	Baseline	End of Intervention	2-Month Follow Up
Mutuality[a] (*t* test)	0.01	1.47	1.47
Intervention group	1.15 (0.40)	1.33 (0.59)	1.27 (0.54)
Control group	1.15 (0.51)	1.19 (0.42)	1.13 (0.50)
Active coping[b] (*t* test)	–0.37	4.63***	2.92**
Intervention group	1.15 (0.40)	1.65 (0.45)	1.45 (0.43)
Control group	1.18 (0.37)	1.28 (0.38)	1.24 (0.32)
Passive coping[b] (*t* test)	–0.21	–0.65	0.54
Intervention group	1.38 (0.42)	1.24 (0.38)	1.13 (0.39)
Control group	1.39 (0.35)	1.29 (0.32)	1.09 (0.30)
Depressive symptoms[c] (*t* test)	1.15	–4.00***	–3.35**
Intervention group	13.93 (3.45)	9.40 (3.39)	10.19 (2.78)
Control group	13.21 (3.11)	11.96 (3.40)	12.21 (3.56)

Note. Data were based on $N = 112$, intervention group $n = 56$, and control group $n = 56$. Actual sample size at end of intervention was $N = 99$, intervention group $n = 54$, and control group $n = 45$. Actual sample size at 2-month follow up was $N = 82$, intervention group $n = 47$, and control group $n = 35$. Missing data were replaced using the missForest method.
[a] Measured using the 15-item Mutuality Scale. Item scores range from 0 to 4, with higher scores indicating higher levels of mutuality.
[b] Measured using the 20-item Simplified Coping Scale, with item scores ranging from 0 to 3. A higher score in active coping (12 items) indicates better coping, whereas a higher score in passive coping (8 items) indicates poorer coping.
[c] Measured using the 10-item Center for Epidemiologic Studies Depression Scale. Scores range from 0 to 30, with higher scores indicating more depressive symptoms.
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

nurses can maintain caregivers' mental health in countries that lack adequate numbers of primary practitioners.

The Role of the Studied Variables in the Interventional Effect

Mutuality. Mutuality was indicated as a significant, although weak, predictor of caregivers' depressive symptoms in the NLCBI after covariates were adjusted. Cognitive training has been found to enhance caregivers' appraisals on dyadic mutuality (Li, Powers, et al., 2012), and higher mutuality is known to mitigate caregiver depression (Crist et al., 2017; Ostwald et al., 2014). However, there was only a slight increase in caregivers' mutuality in the NLCBI group, which was in line with the finding of Kunik et al. (2017) but contrary to findings of Gonzalez et al. (2014) and Orrell et al. (2017). These differences could be due to the strategic diversity among similar studies: for example, the current NLCBI did not fully focus on mutuality but included it as one aspect of the intervention.

In addition, the small increase in mutuality helped explain its non-mediating role in the effectiveness of the NLCBI, with some contributing factors. PWD in the current study with poor cognitive function may have lost their ability to interact or reciprocate in depth, which could have compromised caregivers' mutuality (Park & Schumacher, 2014). Moreover, caregivers in the current study were moderately depressed, which may have decreased their mutuality compared to caregivers who were not depressed (Shim, Landerman, & Davis, 2011); this could explain caregivers' sluggish response to the NLCBI. Although this finding adds to knowledge of the topic, future investigations could focus the intervention on mutuality among caregivers of PWD with less cognitive impairment, therefore helping better understand the dynamics of mutuality and its roles.

Active Coping. The current study showed that, after adjusting for covariates, enhancement in active coping strongly predicted fewer depressive symptoms among caregivers with NLCBI, a finding supported by similar studies (Cheng et al., 2017; Li et al., 2014). This NLCBI partly dealt with caregivers' perceptions of caregiving demands and cultural barriers, such as the social stigma of dementia and caregivers' reluctance to seek help within the social context of China. Caregivers' reappraisals on the perspectives above may have grown more positive, and their active coping strategies improved, resulting in reduced caregiver

TABLE 4

The Fixed Effects of Mutuality, Active Coping, and Passive Coping on Caregivers' Depressive Symptoms in a Generalized Linear Mixed Model (*N* = 112)

	Depressive Symptoms	
Predictor Variables	**ß**	***p* Value**
Intercept	3.18	0.18
Caregiver gender	1.02	0.044*
Patients' Activities of Daily Living score	0.07	0.009**
Patients' Mini-Mental State Examination score	0.13	0.007**
Group	1.41	0.053
Time[a]		
End of intervention	3.70	0.0001***
2-month follow up	4.93	0.0001***
Group × time	1.12	0.0001***
Mutuality	–0.75	0.049*
Active coping	–2.06	0.0001***
Passive coping	1.43	0.001**

Note. Independent variables of mutuality, active coping, and passive coping, and the dependent variable of depressive symptoms, are the reconstructed variables with their baseline, end of intervention, and 2-month follow up scores combined.
[a] Baseline time was used as reference group.
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

depression (Lazarus & Folkman, 1984). Thus, regular, tailored mental health nursing interventions such as this NLCBI could be useful in relieving caregivers' mental distress in developing countries.

However, active coping was not found to play a mediating role in the interventional effect in the current study, which contradicted findings of Gallagher-Thompson et al. (2008) but partially aligned with Li et al.'s (2014) report. The current study did not include the mediating variable of skill utilization as in previous studies, and emotional coping was not measured separately from behavioral coping; however, coping strategies included in these mediators could be reflected in the scale of active coping in the current study. Varied research designs including the use of different scales may have contributed to inconsistencies among these studies. Thus, the mediating role of active coping could be clarified by repeating this study.

Passive Coping. The decrease in passive coping predicted fewer depressive symptoms among caregivers receiving the NLCBI, with covariates adjusted. As shown by López-Martinez (2008), caregivers who received psychosocial interventions perceived higher social support, and, as a result, responded with fewer passive coping strategies. Nevertheless, the decrease in passive coping in this study was contrary to the finding from a similar study by Li et al. (2014) but consistent with Losada et al.'s (2015) finding. These different results could be due to caregiver responses from diversified cultures, or the use of mixed scales: passive coping, dysfunctional coping, or emotion-focused coping.

Moreover, changes in passive coping were not likely to play a mediating role in the interventional effect, as the mediating conditions were not met. It is possible that this NLCBI focused more on benefit-finding, which mainly trained caregivers to have positive thoughts and, consequently, to use more active coping strategies, which may have had less of an impact on passive coping. That said, it is worth noting that strategies of both groups (intervention and control) may be equally effective in affecting caregivers' passive coping (as indicated by *t* test). Therefore, the current study contributed first-hand knowledge of how a decrease in passive coping alleviated depression through mental health nursing interventions among Chinese caregivers. However, this mechanism deserves further confirmation in the target population due to the few references available.

LIMITATIONS

The current study explored a rarely investigated topic—the predicting and mediating roles of mutuality and coping in the effectiveness of a NLCBI on dementia caregivers in China. However, there were limitations in this study.

First, although missing data were replaced by the missForest method, the high attrition rate in the control group and the age difference between individuals who completed and dropped out of the study may have compromised the final results. Caregivers in this study who were depressed were inclined to discontinue caregiving; younger caregivers in China may have other competing roles such as child rearing and were prone to relocate; and PWD in poor condition often did not live through the three survey stages. Future studies should select respondents who would be more likely to complete the study and develop incentives to retain caregivers.

Second, a no-treatment comparison group was not included in this study. It is possible that part of the

TABLE 5

Analysis of the Mediation of Active Coping (N = 112)

Regression Analysis	Independent Variable	Dependent Variable	β	t Test	p Value
	Modules for T_2T_1 Changes				
1	Intervention group	Change in active coping	0.41	4.58	0.0001***
2	Intervention group	Change in depressive symptoms	–0.48	–5.58	0.0001***
3	Intervention group	Change in depressive symptoms	–0.36	–4.03	0.0001***
	Change in active coping		–0.28	–3.17	0.002**
	Modules for T_3T_1 Changes				
1	Intervention group	Change in active coping	0.27	2.88	0.005**
2	Intervention group	Change in depressive symptoms	–0.39	–4.29	0.0001***
3	Intervention group	Change in depressive symptoms	–0.33	–3.61	0.0001***
	Change in active coping		–0.21	–2.22	0.028*

Note: T_1 = Time 1 (baseline); T_2 = Time 2 (end of intervention); T_3 = Time 3 (2-month follow up). T_2T_1 = T_2 Score – T_1 Score; T_3T_1 = T_3 Score – T_1 Score. All regression models controlled for caregiver gender, patients' Activities of Daily Living score, and Mini Mental Status Examination score.
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

interventional effect was due solely to the nurses' attention; in addition, relevant variables such as perceived social support were not included, which may have resulted in biased findings. Therefore, it is suggested that future research designs contain a no-treatment group for observation and a confounding variable of perceived social support.

IMPLICATIONS AND DIRECTION FOR FUTURE RESEARCH

Several implications were derived from the current research. This study partially supported the Stress and Coping Theory by Lazarus and Folkman (1984), as active coping was found to act as a predictor, but not a mediator, in the interventional effect. This NLCBI most likely worked on caregivers' reappraisals, which in turn improved caregivers' active coping skills and mutuality, decreased their passive coping, and ultimately reduced their depressive symptoms.

In relation to nursing practice, this NLCBI, which featured both face-to-face sessions and telephone consultations, was beneficial in reducing the mental stress of depressed, homebound caregivers of PWD. This approach can be further developed, and community nurses should be trained in its use.

As for nursing research, a similar research design could be conducted with a longer observation period while controlling for the attrition rate. Studies aiming to improve caregiver mutuality warrant further exploration, and the mediating role of mutuality or coping needs to be confirmed. Finally, perceived social support can be included as a potential mediator.

China has reached a critical stage in its social policies about dementia care, and it is time to acknowledge family caregivers' contributions. Supporting family caregivers will have a significant effect because caregivers' well-being helps sustain family caregiving to PWD, which in turn greatly reduces the nation's financial health care investments. Thus, mental health nursing services such as NLCBIs should be initiated to meet caregivers' psychological needs. These services may also be combined with other social policies to combat the crisis of dementia care.

CONCLUSION

Improved mutuality and active coping and reduced passive coping played a predicting, but not mediating, role in the NLCBI on relieving depressive symptoms among family caregivers of PWD. This study contributed to preliminary findings on the mechanisms of the intervention. Mental health nursing programs and supportive social policies are suggested to improve caregivers' mutuality and active coping, and to reduce passive coping to maintain caregivers' mental health in countries with similar socioeconomic contexts to China.

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